Problem 1

Part (a)

The function $F(x, y)$ **is not** a distribution function on $\mathbb{R}^2$ because it violates the n-monotonicity condition (Lemma 1.4.9c). Consider the rectangle $(0, 1] \times (0, 1]$. The value assigned to this rectangle is given by

$$\begin{aligned}
\Delta_1 \Delta_1 F(0,0) &= F(1,1) - F(0,1) - F(1,0) + F(0,0) \\
&= (1 - e^{-2}) - (1 - e^{-1}) - (1 - e^{-1}) + 0 \\
&= 1 - e^{-2} - 1 + e^{-1} - 1 + e^{-1} \\
&= -1 + 2e^{-1} - e^{-2} \\
&= -(1 - e^{-1})^2 < 0
\end{aligned}$$

Since F assigns a negative value to a valid Borel set, it is not a valid distribution function.

Part (b)

We show that $F(x, y)$ is the cumulative distribution function generated by a valid probability density $f(x, y)$. If there exists a Borel-measurable function $f : \mathbb{R}^2 \to \mathbb{R}_+$ such that $\int_{\mathbb{R}^2} f(x, y) dx dy = 1$ then $f(x, y)$ is a valid probability density (Definition 1.5.12). If integrating $f(x, y)$ over $(-\infty, x] \times (-\infty, y]$ recovers $F(x, y)$, then $F(x, y)$ is strictly a valid distribution function on $\mathbb{R}^2$.

Step 1. Calculate the mixed partial derivative (candidate density) $f(x, y) = \frac{\partial^2 F}{\partial x \partial y}$ almost everywhere:

1. If $0 < x < y$, then:
$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}(1 - e^{-x} - xe^{-y}) = e^{-x} - e^{-y}$$
$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial}{\partial y}(e^{-x} - e^{-y}) = e^{-y}$$

2. If $0 < y < x$, then:
$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}(1 - e^{-y} - ye^{-y}) = 0$$
$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial}{\partial y}(0) = 0$$

3. If $x < 0$ or $y < 0$, then $F(x, y) = 0$ and thus $f(x, y) = 0$.

In general, we have:

$$f(x, y) = \begin{cases} e^{-y} & \text{if } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

Step 2. Verify that $f(x, y)$ is a valid density:

1. **Non-negativity:** $e^{-y} > 0$ for all $y \in \mathbb{R}$, thus $f(x, y) \geq 0, \forall (x, y) \in \mathbb{R}^2$.

2. **Borel-measurability:** The function f is a continuous function on the open set $\{(x, y) \in \mathbb{R}^2 : 0 < x < y\}$ and is zero elsewhere. Thus, f is Borel-measurable.

3. **Normalization:** We integrate over the support region $0 < x < y < \infty$:

$$\begin{aligned}
\int_0^\infty \int_0^y e^{-y}\,dx\,dy &= \int_0^\infty \left[xe^{-y}\right]_{x=0}^{x=y} dy \\
&= \int_0^\infty ye^{-y}\,dy \\
&\overset{(u=y,\,dv=e^{-y}\,dy)}{=} \left[-ye^{-y}\right]_0^\infty - \int_0^\infty -e^{-y}\,dy \\
&= 0 - \left[e^{-y}\right]_0^\infty = 1
\end{aligned}$$

By Theorem 1.5.11(b), any Borel-measurable function $f : \mathbb{R}^2 \to \mathbb{R}_+$ with $\int_{\mathbb{R}^2} f(x,y)dxdy = 1$ uniquely defines a probability measure $\mathbb{P}$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$.

Step 3. Verify that integrating $f(u,v)$ recovers $F(x,y)$:
Case 1 ($0 \le x \le y$):

$$\begin{aligned}
\int_{-\infty}^x \int_{-\infty}^y f(u,v)dvdu &= \int_0^x \left(\int_u^y e^{-v}dv\right) du \\
&= \int_0^x \left[-e^{-v}\right]_{v=u}^{v=y} du \\
&= \int_0^x (e^{-u} - e^{-y})du \\
&= \left[-e^{-u} - ue^{-y}\right]_{u=0}^{u=x} \\
&= (-e^{-x} - xe^{-y}) - (-1-0) = 1 - e^{-x} - xe^{-y}
\end{aligned}$$

This matches $F(x,y)$ for $0 \le x \le y$.
Case 2 ($0 \le y \le x$):

$$\begin{aligned}
\int_{-\infty}^x \int_{-\infty}^y f(u,v)dvdu &= \int_0^y \left(\int_u^y e^{-v}dv\right) du \\
&= \int_0^y (e^{-u} - e^{-y})du \\
&= \left[-e^{-u} - ue^{-y}\right]_{u=0}^{u=y} \\
&= (-e^{-y} - ye^{-y}) - (-1-0) = 1 - e^{-y} - ye^{-y}
\end{aligned}$$

Since we have shown that the cumulative integral of $f(x,y)$ reconstructs $F(x,y)$ (Step 3) and since $f(x,y)$ uniquely defines a probability measure (Step 2), F is strictly a valid distribution function.

Marginal distributions

- *Marginal distribution of X:* For $x \ge 0$, as $y \to \infty$, the condition $y \ge x$ is eventually satisfied. We use the first branch:

$$F_X(x) = \lim_{y\to\infty} F(x,y) = \lim_{y\to\infty} \left(1 - e^{-x} - xe^{-y}\right) = 1 - e^{-x} - x(0) = 1 - e^{-x}$$

  Thus:

$$F_X(x) = \begin{cases} 1 - e^{-x} & \text{if } x \ge 0, \\ 0 & \text{otherwise.} \end{cases}$$

- *Marginal distribution of Y:* For $y \geq 0$, as $x \to \infty$, the condition $x \geq y$ is eventually satisfied. We use the second branch:

$$F_Y(y) = \lim_{x\to\infty} F(x,y) = \lim_{x\to\infty} \left(1 - e^{-y} - ye^{-y}\right) = 1 - e^{-y} - ye^{-y}$$

  Thus:

$$F_Y(y) = \begin{cases} 1 - e^{-y} - ye^{-y} & \text{if } y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Part (c)

Let $F_1(x,y) = (x \wedge 1)\mathbb{I}_{[0,\infty)}(x) \cdot (y \wedge 1)\mathbb{I}_{[0,\infty)}(y)$ and $F_2(x,y) = \mathbb{I}_{[0,\infty)}(x) \cdot \mathbb{I}_{[1,\infty)}(y)$, so that $F(x,y) = \frac{1}{2}F_1(x,y) + \frac{1}{2}F_2(x,y)$. We prove that both F_1 and F_2 are valid distribution functions on $\mathbb{R}^2$:

- $F_1(x,y)$*:* Let $G_1(x) = (x \wedge 1)\mathbb{I}_{[0,\infty)}(x)$ and $H_1(y) = (y \wedge 1)\mathbb{I}_{[0,\infty)}(y)$. G_1 and H_1 corresponds to the distribution functions of the uniform distribution on $[0,1]$. Since the product of two distribution functions is a distribution function (Example 1.4.13(ii)), then $F_1(x,y) = G_1(x)H_1(y)$ is a valid distribution function on $\mathbb{R}^2$.
- $F_2(x,y)$*:* Let $G_2(x) = \mathbb{I}_{[0,\infty)}(x)$ and $H_2(y) = \mathbb{I}_{[1,\infty)}(y)$. G_2 and H_2 corresponds to the distribution functions of the Dirac measures δ_0 and δ_1 at 1 (δ_1), respectively. Then, the product $F_2(x,y) = G_2(x)H_2(y)$ is also a valid distribution function on $\mathbb{R}^2$.

Let $\alpha = 1/2$, then $F(x,y) = \alpha F_1(x,y) + (1-\alpha)F_2(x,y)$ is a strict convex combination of two valid distribution functions and therefore $F(x,y)$ **is a valid distribution function** on $\mathbb{R}^2$.

Marginal distributions

- *Marginal distribution of X:*

$$\begin{aligned} F_X(x) &= \lim_{y\to\infty} F(x,y) \\ &= \frac{1}{2}(x \wedge 1)\mathbb{I}_{[0,\infty)}(x)\left(\lim_{y\to\infty}(y \wedge 1)\mathbb{I}_{[0,\infty)}(y)\right) + \frac{1}{2}\mathbb{I}_{[0,\infty)}(x)\left(\lim_{y\to\infty}\mathbb{I}_{[1,\infty)}(y)\right) \\ &= \frac{1}{2}(x \wedge 1)\mathbb{I}_{[0,\infty)}(x)(1) + \frac{1}{2}\mathbb{I}_{[0,\infty)}(x)(1) \\ &= \frac{1}{2}\mathbb{I}_{[0,\infty)}(x)\left((x \wedge 1) + 1\right) \end{aligned}$$

  Writing this out explicitly:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0, \\ \frac{1}{2}(x+1) & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x \geq 1. \end{cases}$$

- *Marginal distribution of Y:*

$$\begin{aligned} F_Y(y) &= \lim_{x\to\infty} F(x,y) \\ &= \frac{1}{2}\left(\lim_{x\to\infty}(x \wedge 1)\mathbb{I}_{[0,\infty)}(x)\right)(y \wedge 1)\mathbb{I}_{[0,\infty)}(y) + \frac{1}{2}\left(\lim_{x\to\infty}\mathbb{I}_{[0,\infty)}(x)\right)\mathbb{I}_{[1,\infty)}(y) \\ &= \frac{1}{2}(1)(y \wedge 1)\mathbb{I}_{[0,\infty)}(y) + \frac{1}{2}(1)\mathbb{I}_{[1,\infty)}(y) \\ &= \frac{1}{2}(y \wedge 1)\mathbb{I}_{[0,\infty)}(y) + \frac{1}{2}\mathbb{I}_{[1,\infty)}(y) \end{aligned}$$

Writing this out explicitly:

$$F_Y(y) = \begin{cases} 0 & \text{if } y < 0, \\ \frac{1}{2}y & \text{if } 0 \le y < 1, \\ 1 & \text{if } y \ge 1. \end{cases}$$

Part (d)

The function $F(x, y)$ **is not** a distribution function on $\mathbb{R}^2$ because it violates the necessary condition that $\lim_{x\to-\infty} F(x, y) = 0$ for all $y \in \mathbb{R}$ (Lemma 1.4.9a). This limit yields:

$$\begin{aligned} \lim_{x\to-\infty} F(x, y) &= \lim_{x\to-\infty} \left(\mathbb{I}_{\{x\ge1\}} + \mathbb{I}_{\{y\ge1\}} - \mathbb{I}_{\{x\ge1,y\ge1\}}\right) \\ &= 0 + \mathbb{I}_{\{y\ge1\}} - 0 \\ &= \mathbb{I}_{\{y\ge1\}} \end{aligned}$$

For any $y \ge 1$, this evaluates to:

$$\lim_{x\to-\infty} F(x, y) = 1 \ne 0$$